Object-Centric Grounding of Reasoning in Vision–Language–Action Models for Autonomous Driving

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1 Motivation and Research Gap

Vision–language–action (VLA) models have recently been proposed as a route to interpretable end-to-end driving. Alpamayo-R1 (AR1) [1] is representative: a vision–language backbone generates a short Chain-of-Causation (CoC) reasoning trace, and a flow-matching action expert decodes a trajectory conditioned on that trace. The reported gains are concentrated exactly where imitation learning is weakest: 12% improvement in planning accuracy on challenging cases and a 35% reduction in close-encounter rate in closed-loop simulation.

The reasoning trace is therefore not decorative: it is a functional intermediate representation between perception and planning. End-to-end driving removed the hand-crafted interface between those stages. AR1 restores one, but expresses it in *natural language*.

This creates a structural weakness. A CoC trace refers to agents in textual representation - *"the black SUV from the left is going straight"* - but no component of the architecture requires that agent to exist. The image tokens the reasoning conditions on are shaped solely by the language-modelling objective: AR1 has no detection head, no depth or segmentation supervision, and no object-level entry anywhere in the KV-cache that **the reasoning and the action expert read**. The authors list *weak visual grounding* among the limitations of supervised fine-tuning, noting that next-token prediction does not enforce visual consistency and that the model may hallucinate causal factors not present in the scene. Their mitigation is post-hoc: during RL post-training, a separate large reasoning model grades generated traces. This applies statistical pressure during training but leaves nothing in the architecture constraining what the reasoning may claim, and nothing at all at inference time. **Their own future work accordingly calls for auxiliary objectives** - depth estimation, scene flow, or 3D Gaussian Splatting, to strengthen the visual backbone.

A second, related gap is temporal. The CoC labelling protocol enforces *causal locality*: every causal factor must originate in the observed 2 s history window. Combined with a KV-cache that is rebuilt each frame, this means the model re-derives the scene from scratch at every inference step and holds no persistent identity for any agent. Under occlusion it can state that a vehicle blocks the view, but it cannot state that a previously tracked cyclist is likely still behind it.

Research Gap: Current driving VLAs couple perception to reasoning through an *unverifiable, memoryless interface*. The representation the reasoning refers to is neither grounded in sensor evidence nor persistent across time.

2 Research Objectives and Hypotheses

This proposal introduces an explicit, sensor-grounded, temporally propagated object representation into the VLA sequence, and evaluates whether it improves the fidelity of the generated reasoning and the quality of the resulting trajectories. Three falsifiable hypotheses:

H1 (Grounding failure is measurable and non-negligible.)
Traces generated by the released AR1 model refer to object categories absent from the scene at a rate significantly above that of the human-verified CoC annotations on the same clips.

H2 (An object-centric span reduces grounding failure.)
Conditioning the reasoning on an explicit span of detection-derived agent queries reduces the hallucination rate of H1 without degrading open-loop planning accuracy (minADE_6) or closed-loop safety metrics.

H3 (Temporal propagation extends the benefit beyond the observation window.)
Propagating agent queries across frames, rather than re-detecting them independently, improves reasoning

fidelity and trajectory quality specifically on occlusion and reappearance scenarios, relative to the stateless variant of H2.

A negative result for H1 is informative and cheap to obtain, and the work plan is sequenced so that it is established before any architectural commitment is made.

3 Preliminary Work

My Master's thesis implements a query-based perception-and-prediction pipeline on nuScenes [10], following the agent-query line of DETR3D [6], MOTR [4], MUTR3D [5] and ViP3D [3]: object queries are initialised from a detection heatmap in the manner of TransFusion [7], propagated across frames so that identity is carried by the query itself, and consumed directly by the prediction head without an intermediate box or trajectory interface. The two stages refer to the training schedule: Stage 1 trains the detection and query-propagation backbone on LiDAR input alone, and Stage 2 initialises from the Stage 1 checkpoint and adds spatially modulated co-attention [8] to the decoder cross-attention. Rows are therefore comparable within a stage at matched query budget and batch size, and across stages at fixed query count. Table 1 reports a subset of the ablation study.

Configuration	Queries	Batch	SMCA	EPA↑	minADE↓	minFDE↓	MR↓
Stage 1, 100 queries	100	8	–	0.256	1.022	1.937	0.239
Stage 1, 300 queries	300	16	–	0.277	0.997	1.810	0.177
Stage 1, 300 queries	300	8	–	0.293	0.956	1.776	0.183
Stage 2, 300 queries	300	16	✓	0.295	0.917	1.760	0.175
Stage 2, 300 queries	300	8	✓	**0.318**	**0.882**	**1.590**	0.176
ViP3D [3], camera-only (DETR3D backbone)				0.236	2.03	2.90	0.239
PnPNet-vision [9], camera-only				0.213	2.04	3.03	0.271

Table 1: Ablation on nuScenes. EPA is the end-to-end prediction accuracy of [3].

Two findings bear directly on this proposal. First, query count matters and saturates: raising the budget from 100 to 300 queries improves EPA by 3.7 points at fixed batch size, which indicates that a compact set in the low hundreds is sufficient - relevant because any span inserted into a VLA sequence competes for a token budget that AR1 deliberately minimises. Second, adding spatially modulated co-attention [8] to the decoder cross-attention improved every metric, but at a measurable cost in inference latency. The gain is therefore merely an accuracy–compute trade-off.

4 Methodology and Experimental Design

All work fine-tunes released checkpoints, using the publicly available Alpamayo inference and post-training recipes together with the AlpaSim closed-loop simulator [2].

4.1 Phase 1 — Quantifying grounding failure (H1)

No training is required. The released AR1 checkpoint is run over clips for which 3D object annotations exist, and each generated trace is parsed for referenced object categories and relative positions. A reference is scored as *unsupported* when no annotated object of that category exists in the corresponding region of the scene. The same measurement is applied to the human-written CoC traces for the same clips. Matching free text to detection labels is inherently imprecise, so some references will be flagged as unsupported even when the object was really there; the human traces show how often that happens.

4.2 Phase 2 — An object-centric span (H2)

Agent queries from the perception module are projected into the backbone embedding space and inserted as an explicit span *before* the reasoning tokens:

$$[\, o_{\text{image}},\ o_{\text{egomotion}},\ \mathbf{q}_{1:N},\ \text{Reason},\ \tau\,].$$

Causal attention then exposes every query to every reasoning token, and the KV-cache slice read by the action expert is extended to cover the new span, so that grounding reaches both the language and the control path. The detection loss supervising the queries constitutes the auxiliary objective called for in [1], but targeted at grounding the reasoning rather than at improving features generally.

N is held in the tens, selected by decision relevance rather than raw confidence - consistent with the annotation-economy principle of the CoC protocol, whose traces reference two or three objects - and the token cost is reported explicitly alongside latency.

Controls. Two ablations are built in from the start, since the dominant failure mode is that a model with an existing path to the answer simply ignores a redundant input.

(a) *Span masking*: queries are replaced with learned null embeddings at inference and the change in generated traces is measured.

(b) *Span corruption*: queries from a different scene are substituted; a trace that does not change indicates the span is inert, while one that adopts the foreign objects indicates it is used.

Attention mass placed on query positions by reasoning tokens is logged throughout as a continuous diagnostic.

4.3 Phase 3 — Temporal propagation and a grounding reward (H3)

Queries are propagated between frames with ego-motion compensation, following [4, 5], rather than re-detected independently. Evaluation targets occlusion and reappearance scenarios, where the stateless variant cannot by construction retain an agent that leaves the observation window.

If time permits, the RL post-training stage is extended with a grounding reward analogous to the reasoning–action consistency reward of [1]: parse the generated trace into referenced object categories, compare against the query set, and score the match. The reward is rule-based and requires neither new annotation nor a judge model.

Example 1. Scoring of the proposed grounding reward.

Trace	"yielding to the pedestrian on the right, lead vehicle stopped"
Referenced	{pedestrian, vehicle}
Query set	{car, car, truck} $\rightarrow$ {car, truck}
Score	pedestrian unmatched $\rightarrow$ penalty

4.4 Evaluation protocol

- **Grounding:** unsupported-reference rate (Phase 1 metric), reported per category and per scenario; attention mass on the query span.
- **Open-loop:** minADE_6 and minFDE over the 6.4 s horizon, on nominal and long-tail splits separately, since the claimed benefit is concentrated in the latter.
- **Closed-loop:** close-encounter rate (all and at-fault), off-road rate and simulator score in AlpaSim [2]; scenario-level breakdown for occlusion cases.
- **Cost:** added tokens, added parameters and end-to-end latency, benchmarked against the 99 ms budget reported in [1].

5 Expected Outcomes

1. The first quantitative characterisation of object-level hallucination in a released driving VLA, with a reusable measurement protocol. This stands independently of the outcome of H2 and H3.
2. An architecture and training recipe for inserting a sensor-grounded, temporally propagated object span into a VLA sequence, with ablations establishing whether the span is used rather than ignored.
3. A characterisation of the token-budget/grounding trade-off, directly relevant to deployment on automotive-grade embedded hardware.
4. A workshop or conference submission, with code released against the public Alpamayo checkpoints and AlpaSim scenario sets to support reproduction.

6 Work Plan

The stay is organised as three phases, each resolving one hypothesis and each producing a result that stands on its own, preceded by a month of environment setup and baseline reproduction. Phase 1 is scheduled first because it requires no training and resolves whether the premise holds, so a negative result there could redirect the remaining effort. It also serves as a feasibility check on the data, since the 3D annotations it requires on clips that carry CoC traces are the same annotations needed to train the detector in domain. Phase 2

necessarily precedes Phase 3, because the stateless span is the control against which temporal propagation is measured. The grounding reward is marked optional and is the first item to be dropped, if time won't permit it.

Period	Focus	Deliverable
February	Environment, checkpoints, data; reproduce reported AR1 baselines	Reproduction report
February–March	Phase 1: grounding measurement protocol and results	H1 resolved; draft Section 1
March–April	Phase 2: object-centric span, masking and corruption ablations	H2 resolved; open-loop results
April–May	Phase 3: temporal propagation; closed-loop evaluation in AlpaSim	H3 resolved
May–June	Grounding reward (if feasible); write-up and code release	Manuscript draft

7 Relation to the Activities of the Mobile Perception Systems Lab

The lab's stated long-term goal is spatio-temporal reasoning that allows a mobile platform to predict future events, identified as a key bottleneck for autonomous operation in crowded inner-city settings. Hypothesis H3, agent queries propagated so that identity survives occlusion and reappearance, is a concrete instance of that problem. The lab's test vehicle carries a 360-degree camera setup with LiDAR and radar, which enables the multimodal variant in which LiDAR-derived queries ground an otherwise camera-only backbone. My preliminary work (Table 1) is LiDAR-based and transfers most directly to that configuration.

The group's current work on end-to-end driving models spans perception, reasoning, simulation and embedded deployment, targeting both the prototype vehicle and the TU/e digital twin. This proposal sits in the perception and reasoning parts of that span and connects to three further lines of its research. Work on multi-scale deformable attention [16] and on token-efficient vision transformers [12, 13] addresses the cost an added sequence span incurs, which the token-budget analysis in Phase 2 measures. Radar perception [15] suggests extending the multimodal variant to conditions where the camera backbone is least reliable. Closed-loop evaluation would run in the digital twin alongside the open AlpaSim scenario sets [2].

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